Building & securing Azure VM for SSH related CTF challenge

1. Create an Azure VM

1.1 configurations

- Region chosen: East Asia
- VM size: Standard B2ats v2 (2 vcpus, 1 GiB memory)
- OS installed: Linux (ubuntu server 24.04)
- VM architecture: x64 (others available were: AMD and ARM64)

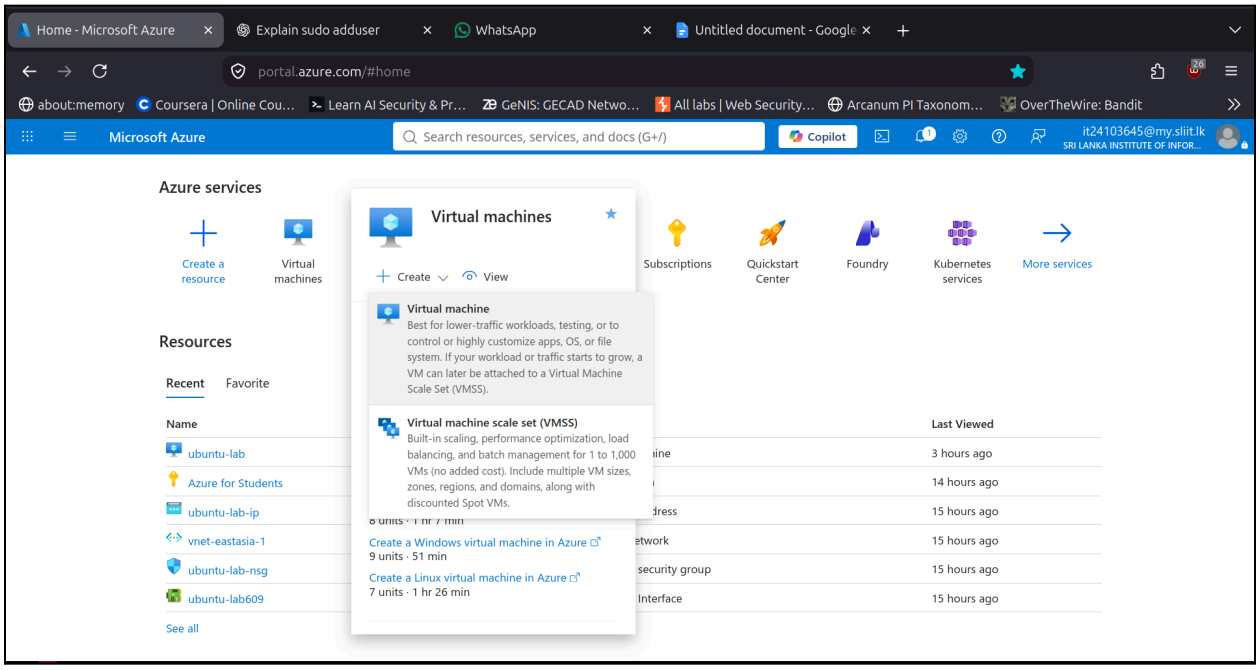


Figure 1: creating a new VM from Azure home dashboard

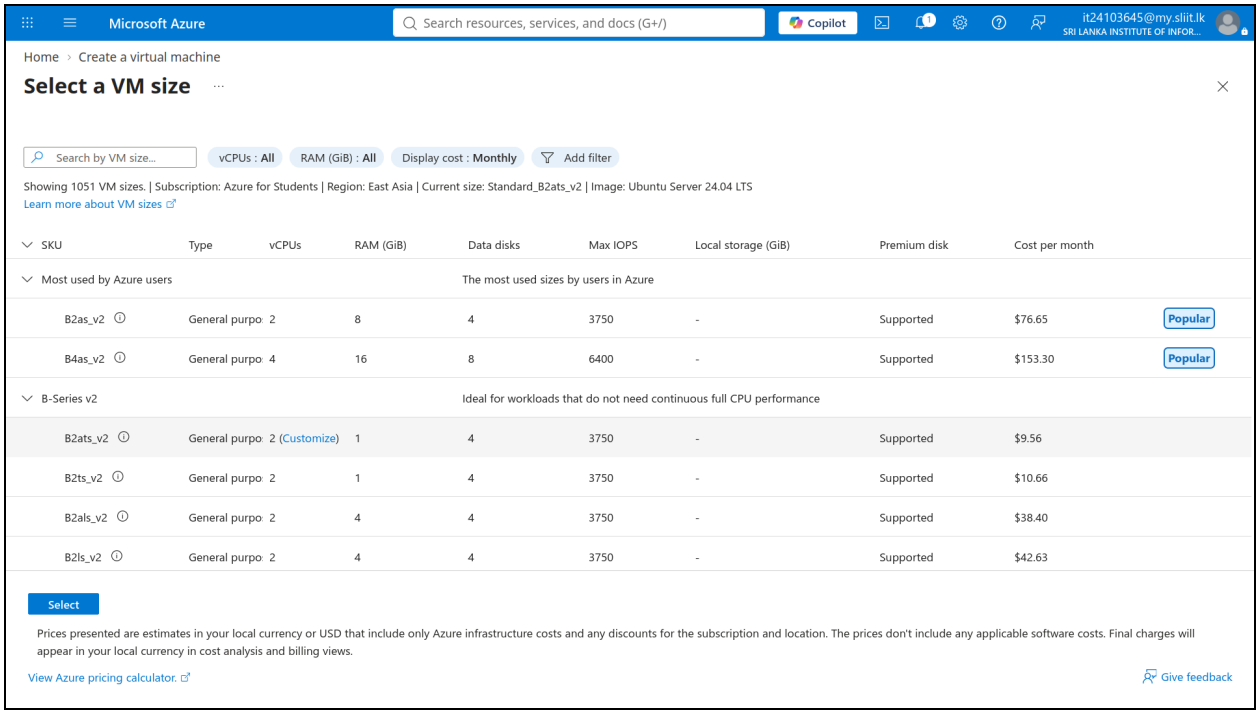


Figure 2: choosing a VM size.

- For East Asia region, only B series V2 VMs are available. B series VMs have Baseline CPU limits (30%) where we are not allowed to exceed the limit for a long period of time

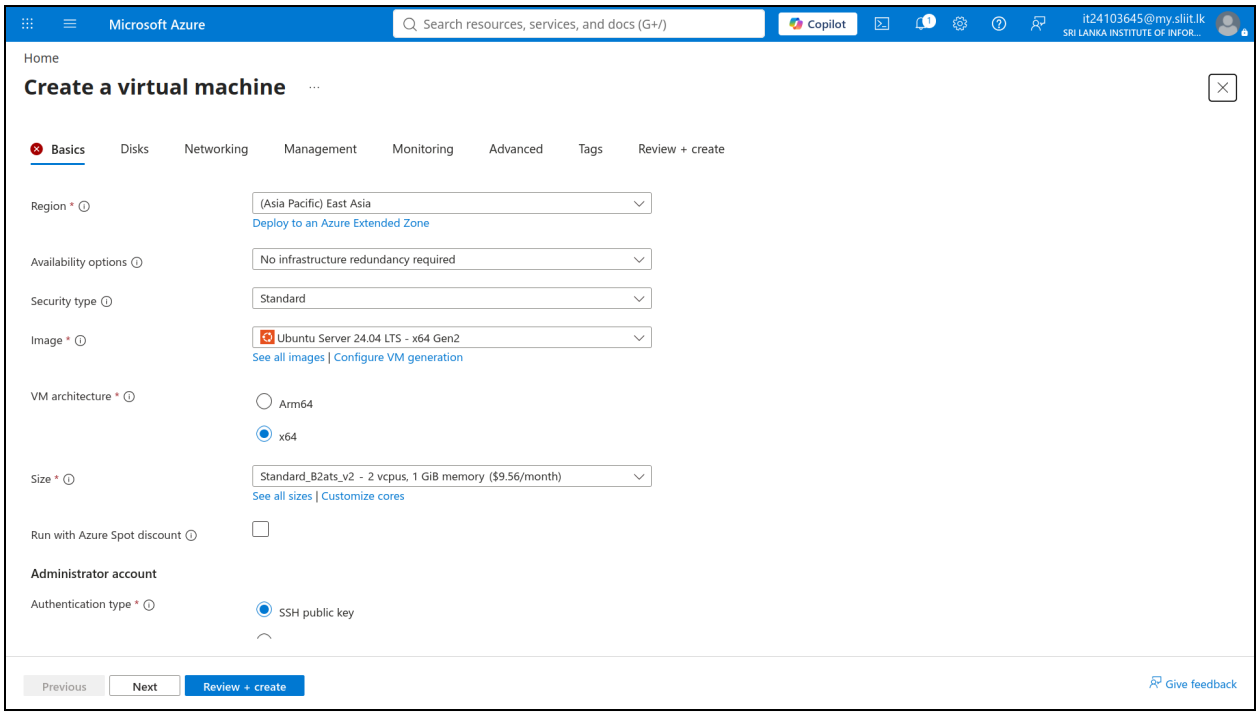


Figure 3: basic configurations

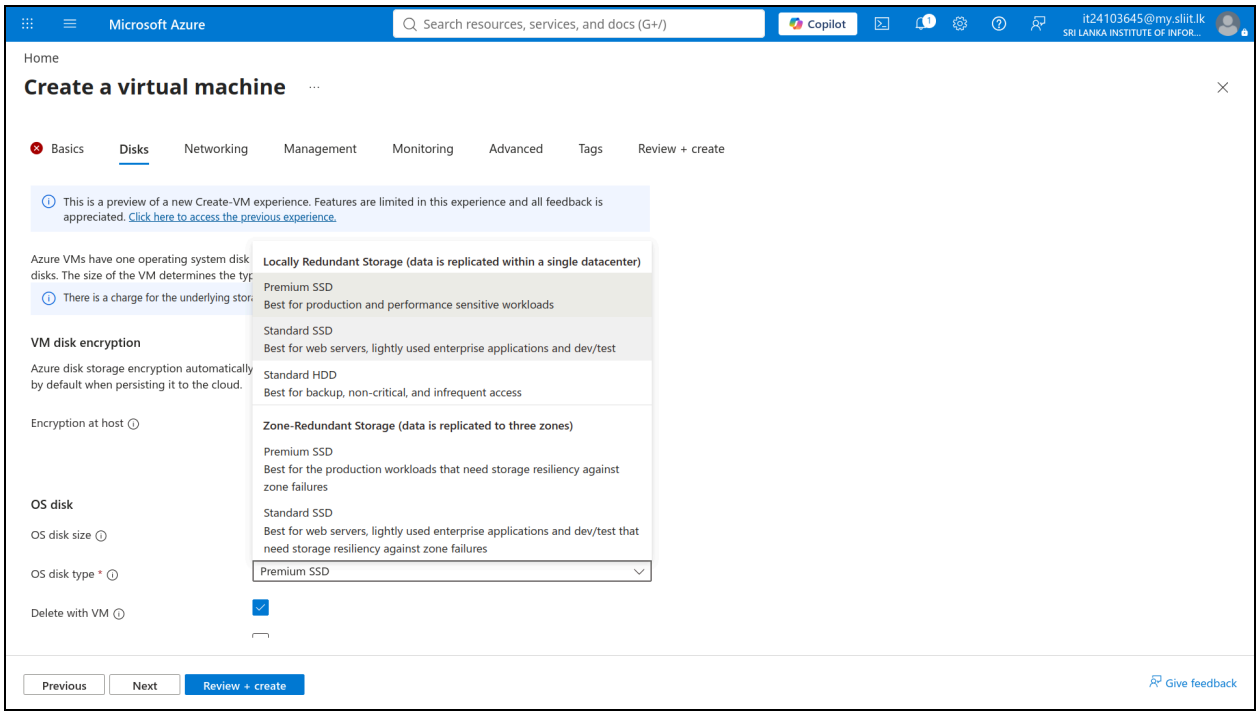


Figure 4: switching from premium SSD to a standard SSD

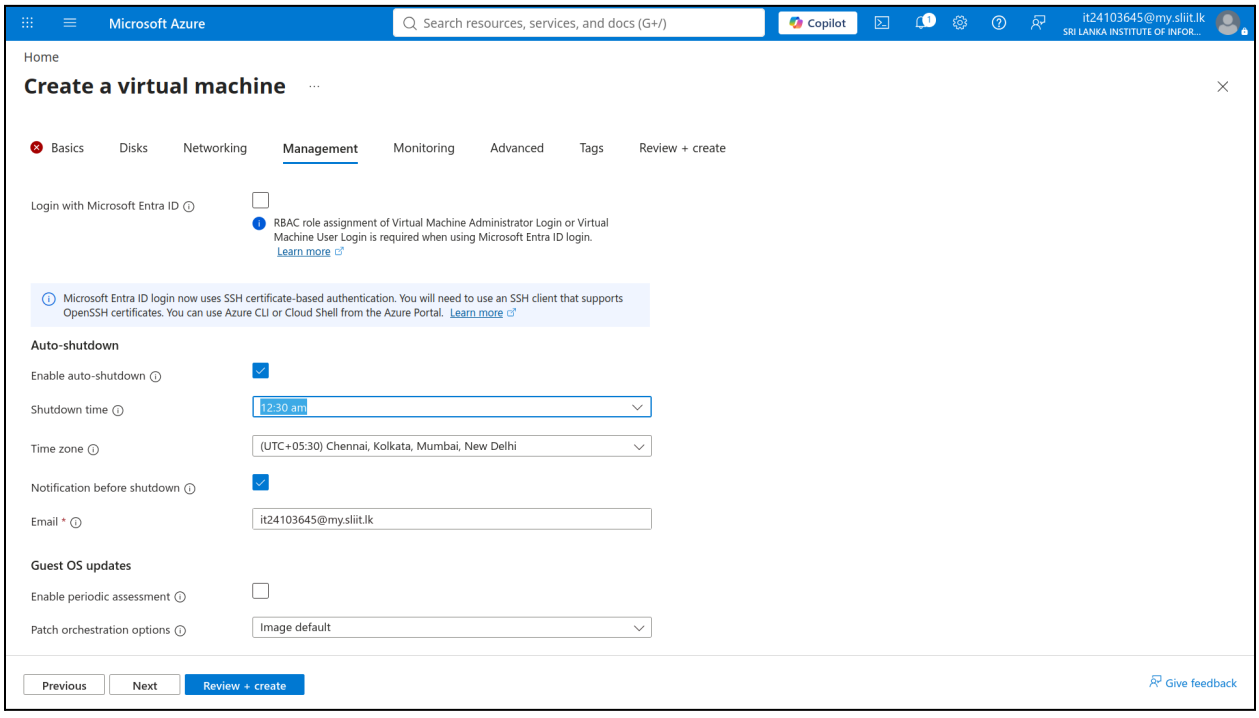


Figure 5: Turning on auto-shutdown at 12:30am everyday.

- Auto-shutdown is a useful security settings to prevent VM running forever in case we forgot to turn the VM off

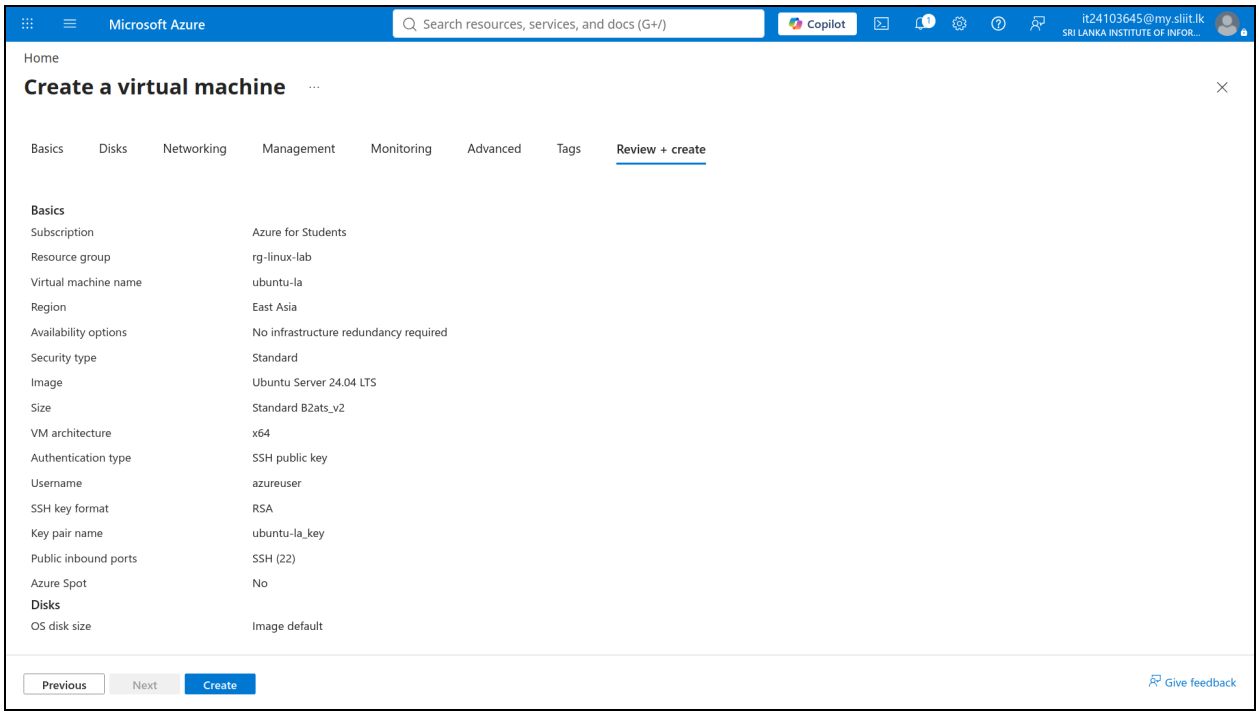


Figure 6: review and then create.

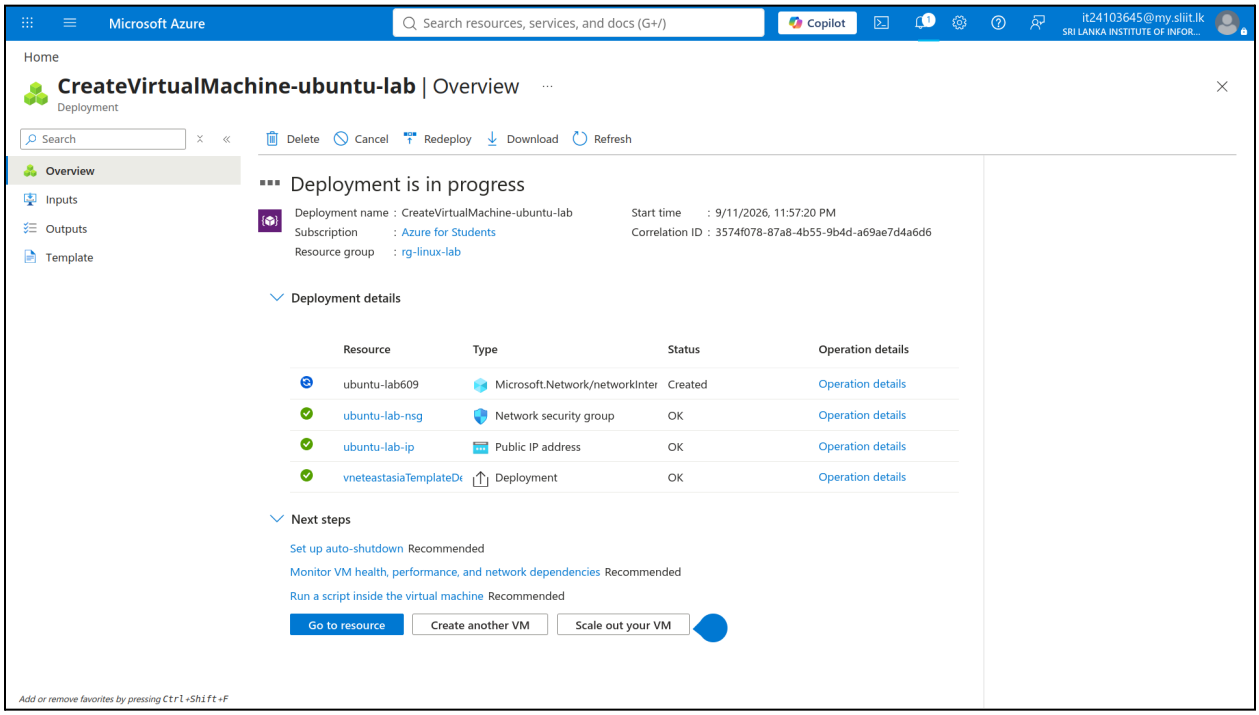


Figure 7: deploying the VM.

1.2 Running the virtual machine (name: ubuntu-lab)

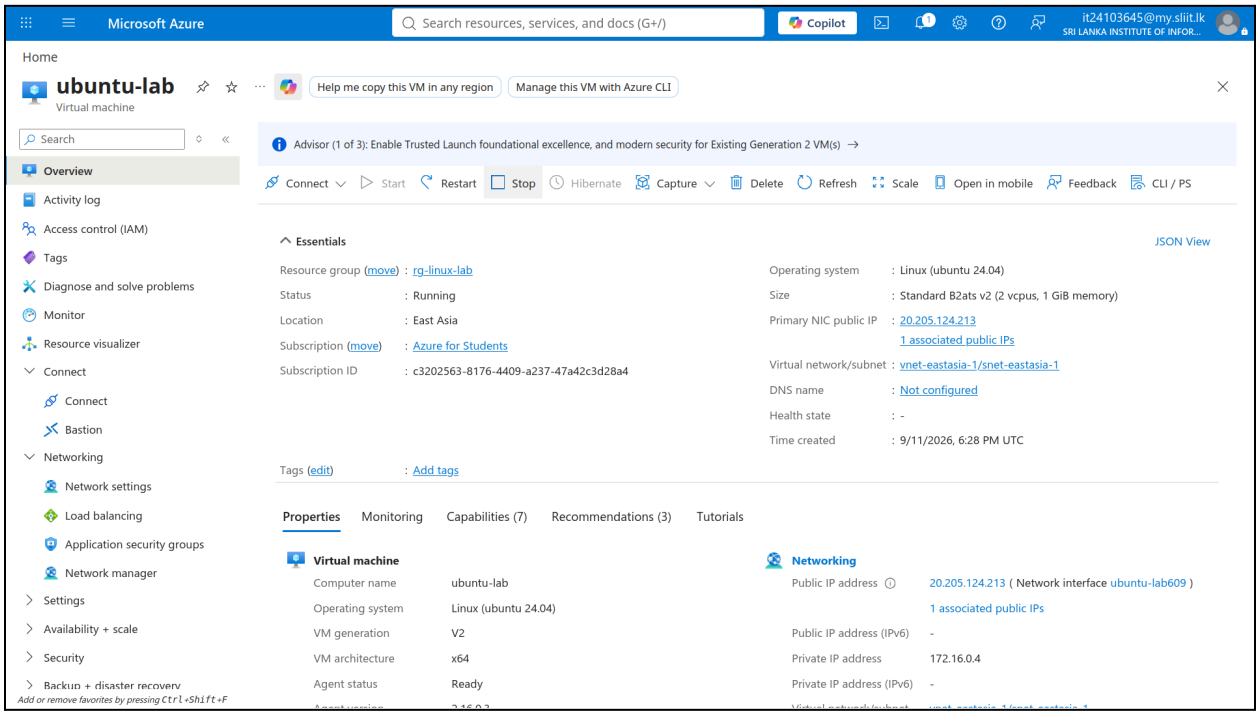


Figure 8: VM control panel.

- Public IP given by Azure: 20.205.124.213

2. SSH into VM with root user account

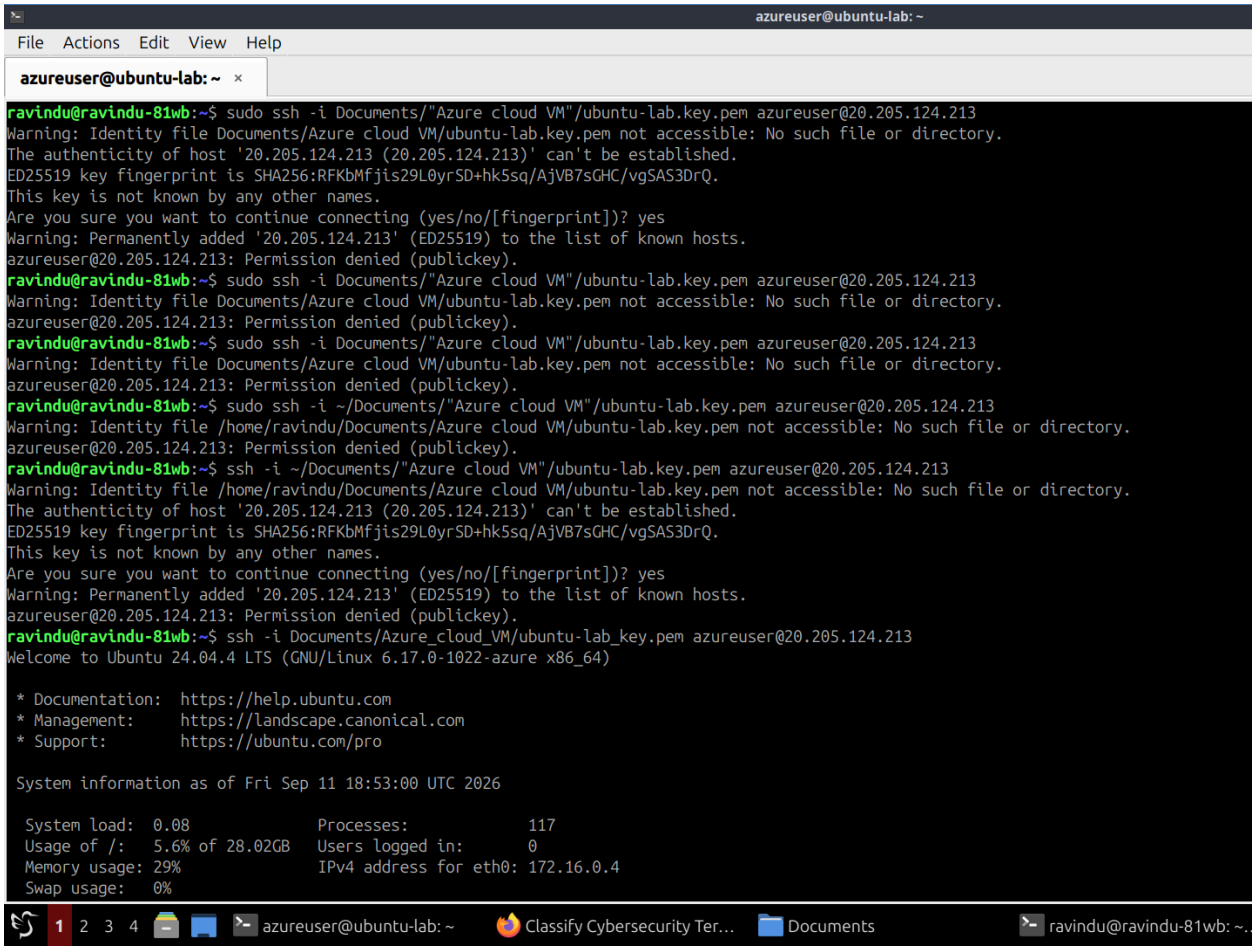


Figure 9: managing to ssh into VM after failing multiple times 😣😣😖

Command used:
ssh -i <path to pem file which has SSH private key> azure_root_username@server_ip

3. Creating a separate user account for CTF players to log into through SSH & securing it

3.1 first make a group named “ctf-lab”

Commands: sudo groupadd ctf-lab

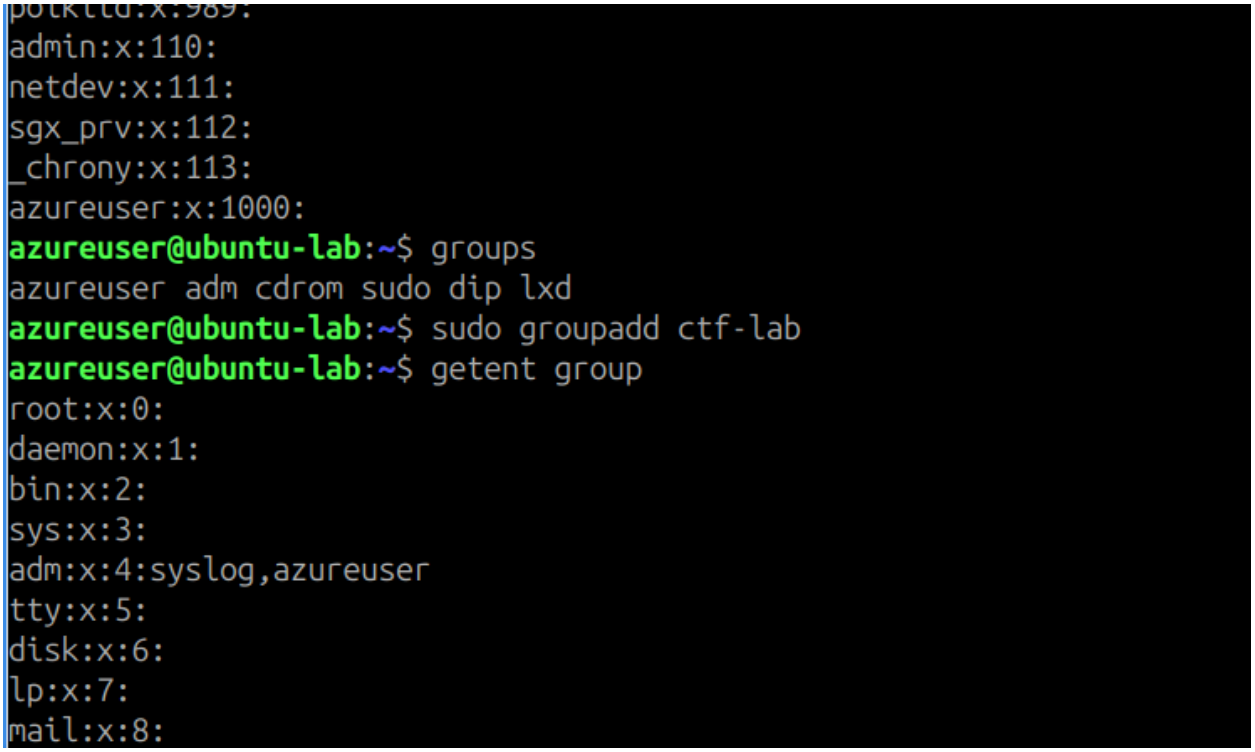


Figure 10: creating a group called “ctf-lab” 😣😣😣

3.2 Then create a user and add it to group

Command used: sudo useradd -g <username> ctf-lab
(Here i have made a user named “cyber-lab” which i later renamed to “ctf-player”)

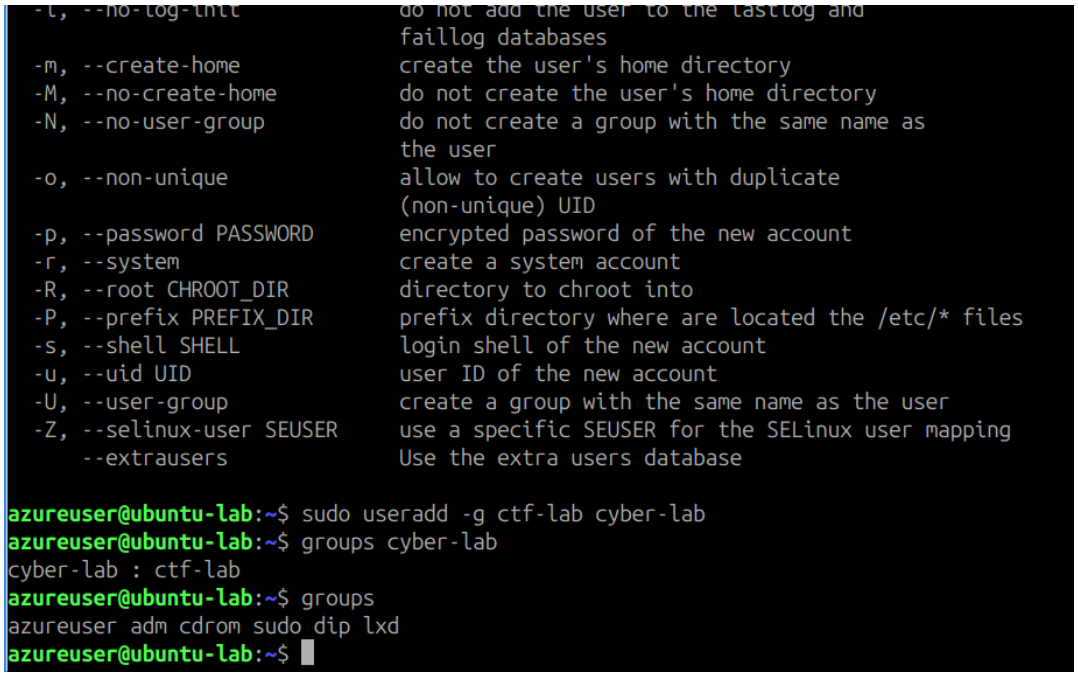


Figure 11: creating a new linux user named “cyber-lab” (later renamed to ctf-player)😣😣😖 and adding the new user immediately into ctf-lab group

We can check the currently available groups in the system with getent group command

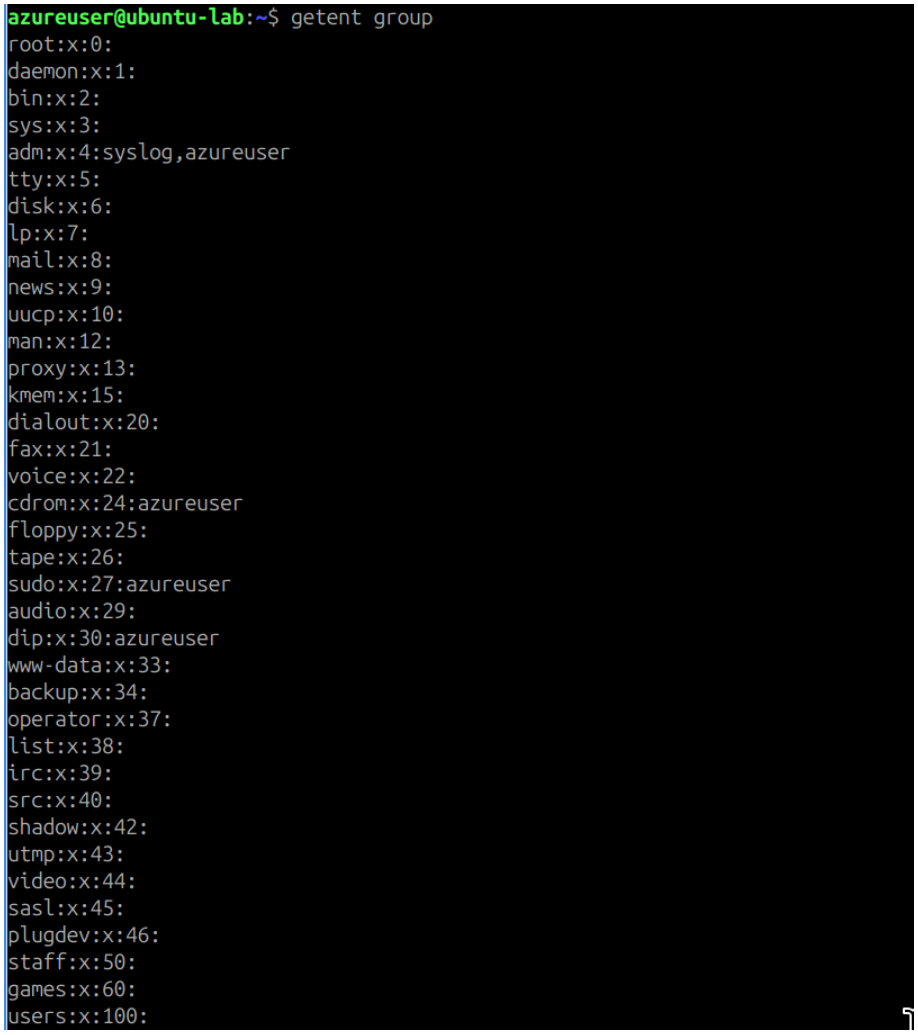


Figure 12: all groups present within the linux VM 😣😣😖

You can see that there's around 30+ groups.This is apparently how linux manages permissions to certain components of OS. Each group gives the users within it a certain set of permissions.

- Ex: users in group cdrom has access to cdrom related stuff
- Ex2: the root user is in multiple groups (adm, cdrom, sudo, dip, lxd) and being in group such as sudo gives the root user the ability to run commands with sudo privileges.

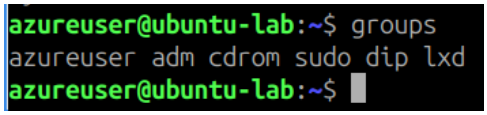


Figure 13: groups that the root user "azureuser" has been added into

```
azureuser adm cdrom sudo dip lxd
```

So your user is a member of **6 groups**:

Group	What it generally means
`azureuser`	Your **primary group** — usually created along with your user
`adm`	Allows access to certain system logs
`cdrom`	Gives access to CD/DVD devices
`sudo`	Allows you to run commands as administrator using `sudo`
`dip`	Related to networking/PPP access
`lxd`	Gives access to the LXD container management system

Figure 14: what some of the groups represents😭😭😭

3.3 Renamed user"cyber-lab" to a more semantically appropriate "ctf-player"

(Username cyber-lab makes no sense. Ctf player has semantic meaning.)
Command used: sudo usermod -l <new name> cyber-lab

```
azureuser:x:1000:
ctf-lab:x:1001:
azureuser@ubuntu-lab:~$ ls
ravinud_was_here.txt
azureuser@ubuntu-lab:~$ sudo usermod -l ctf-player cyber-lab
azureuser@ubuntu-lab:~$ pwd
/home/azureuser
```

Figure 15: renaming username from cyber-lab to ctf-player 😭😭😭

3.4 make a new directory for ctf-player and set it as their home directory.

This is important because when a user logs into SSH, linux gives them a shell. Shells need a home directory or else you'll just see a "$" and your commands will run at the root.

Giving a home directory provides shell with a directory to start shell in.

To make a normal directory into a home directory we have to:

- Set ownership of that directory to the new user
- Give full permissions to owner of that directory

- Assign that directory path as home directory to the new user.

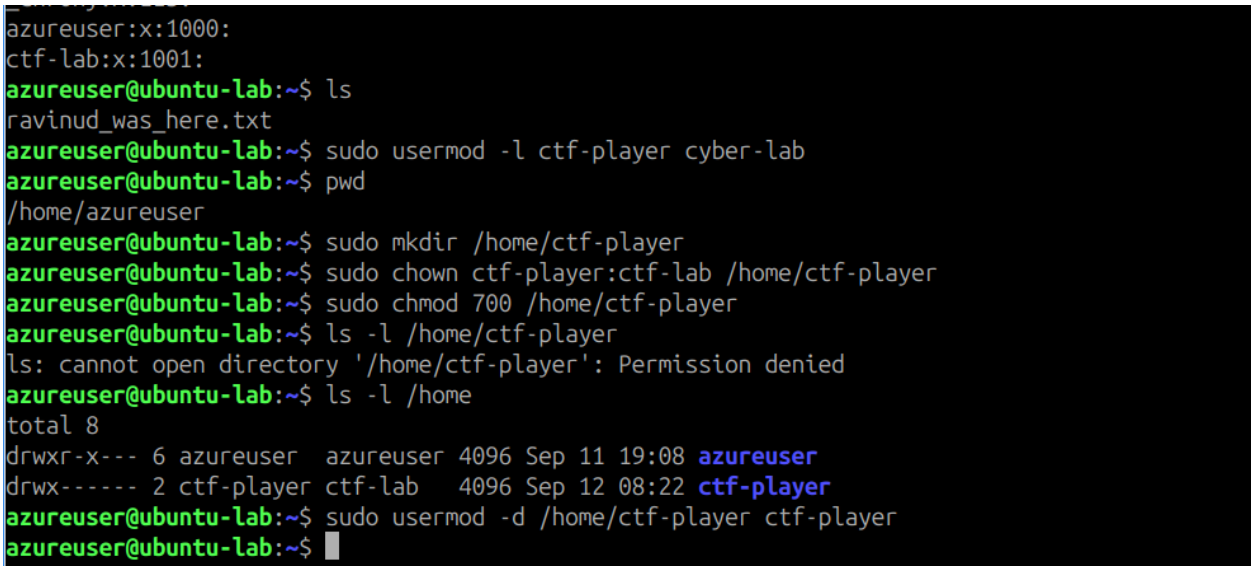


Figure 16: creating a directory called "ctf-player" then adding ownership to user "ctf-player" and giving full permission to owner. 😣😣😖

3.5 Add a pwd for the user as well

This will be used when loggin into the VM through SSH with ctf-player account
Command used: sudo passwd ctf-player

A 25 length, automatically generated password including numbers and special characters were given for prevent bruteforce attacks. Since the Azure IP is open to the public, and the ssh port 22 is also open, a weak password poses a risk.

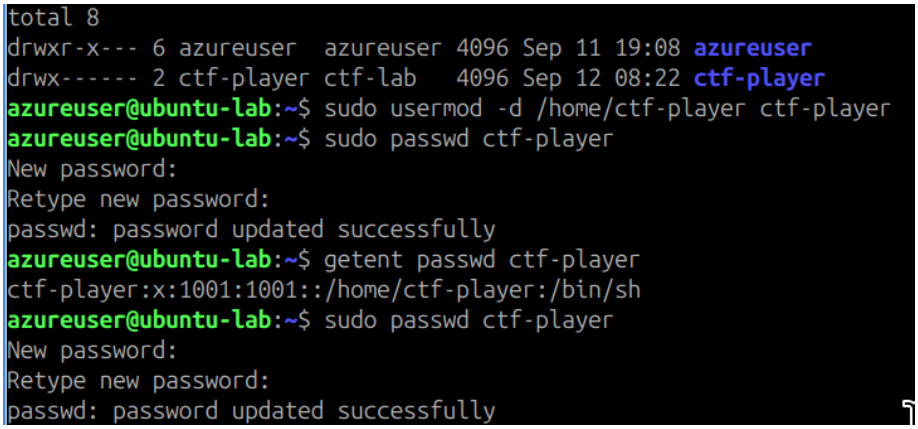


Figure 17: adding a password to ""ctf-player" user account 😣😣😖

3.6 change linux configuration files to allow password login from ssh.

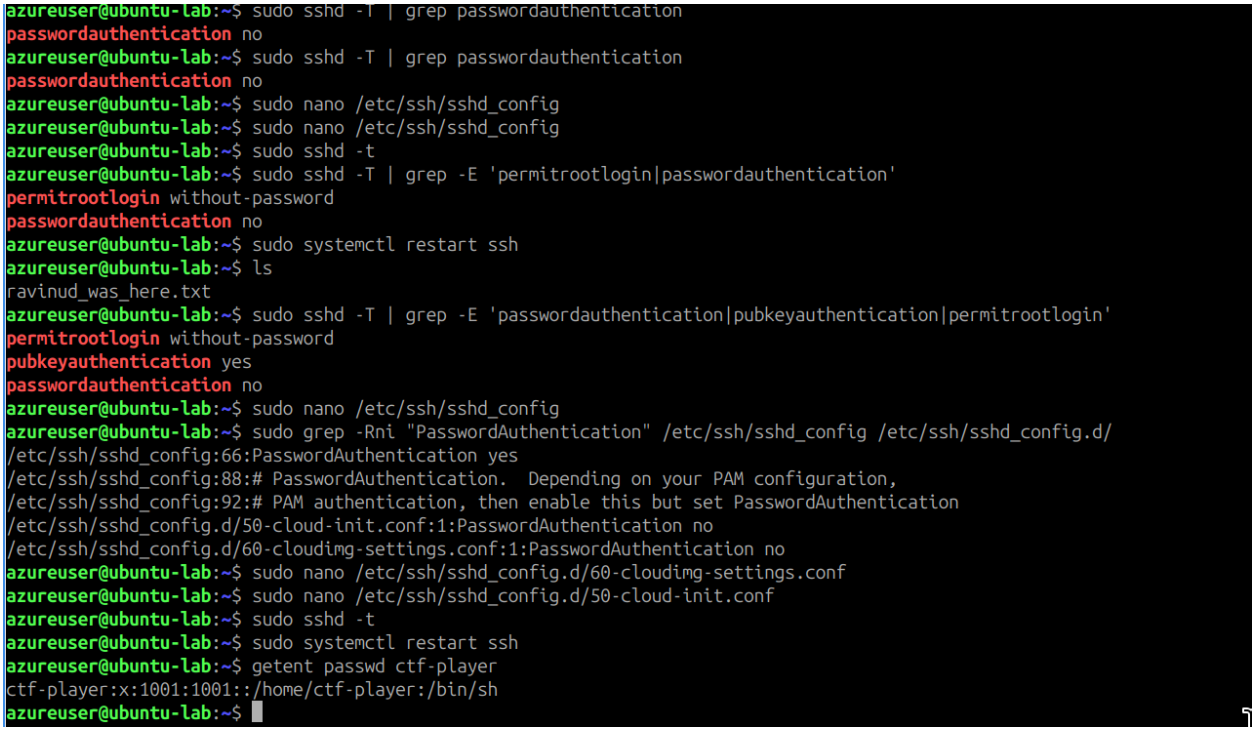


Figure 18: editing the 3 files to allow password authentication except for root user

By default only ssh key pair based logins are allowed.

To change this we have to set “passwordAuthentication yes” in the following files:

- /etc/ssh/sshd_config
- /etc/ssh/sshd_config.d/60-cloudimg-settings.conf
- /etc/ssh/sshd_config.d/50-cloud-init.conf

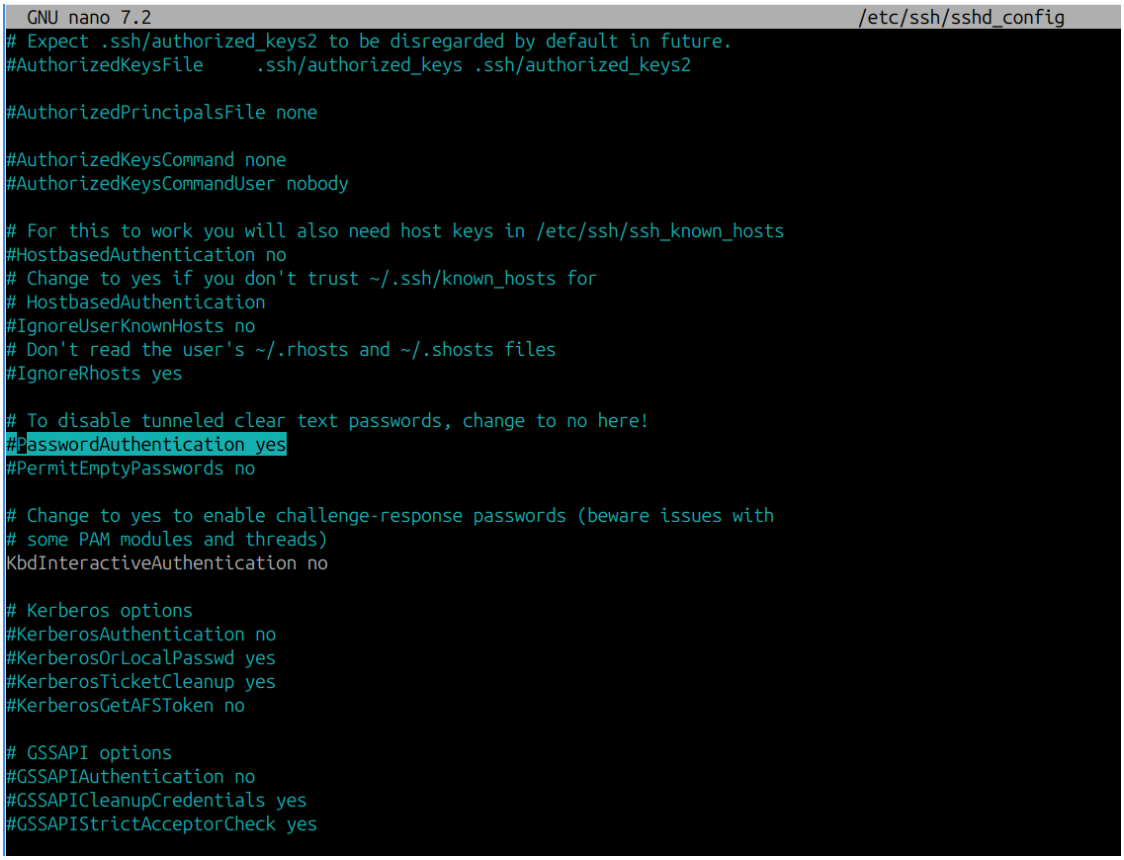


Figure 19: allowing password authentication 😓😓😖

Also uncomment “PermitRootLogin prohibit-password” in this file: /etc/ssh/sshd_config so that root user still have to use the ssh key pair to login.

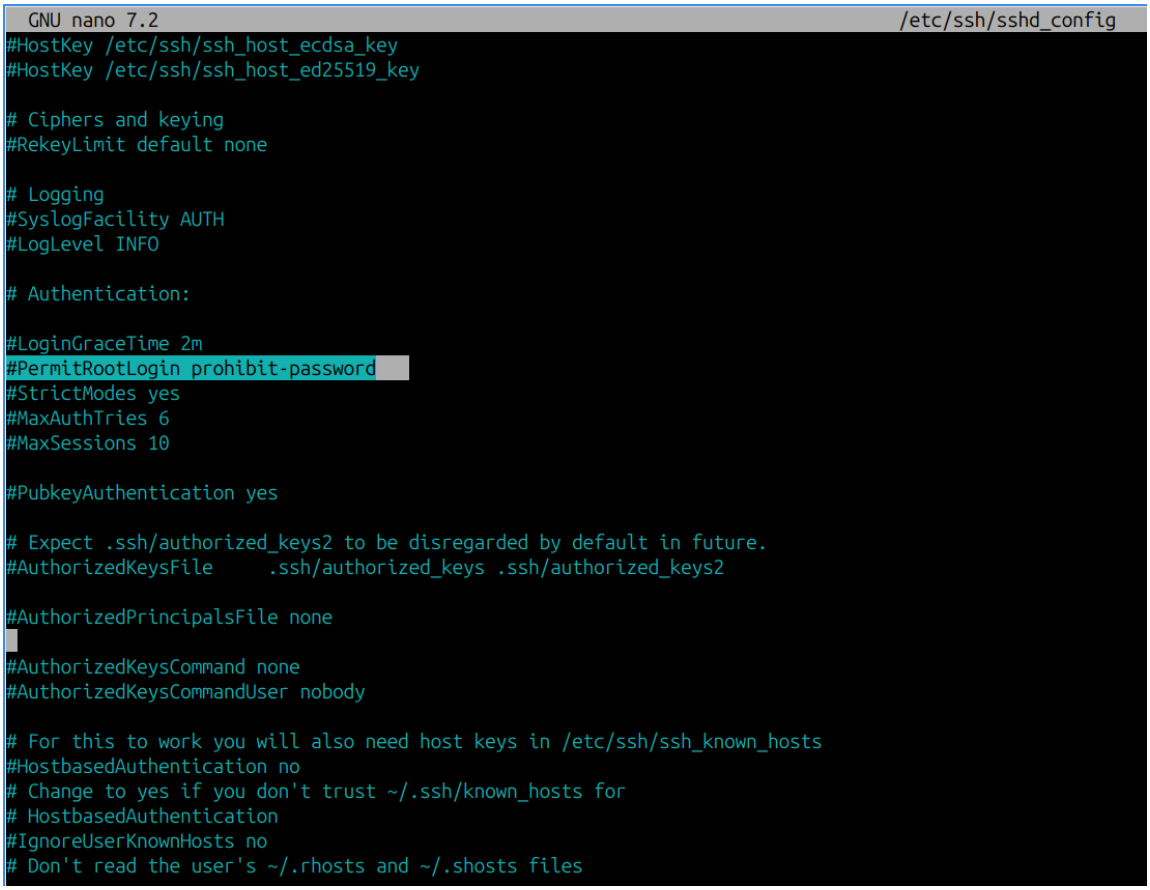


Figure 20: making root user log into ssh with SSH key pair.

3.7 Finally change the CTF player's login shell to "bin/bash" from "bin/sh"

Command used: sudo usermod -s /bin/bash ctf-player

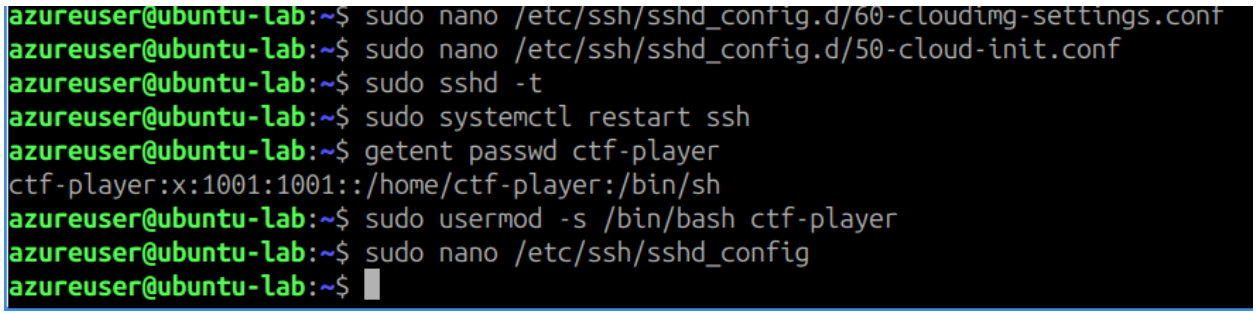


Figure 21: change ctf-player's login shell to bash

3.8 copied these two files into home directory of ctf-player and restarted bash

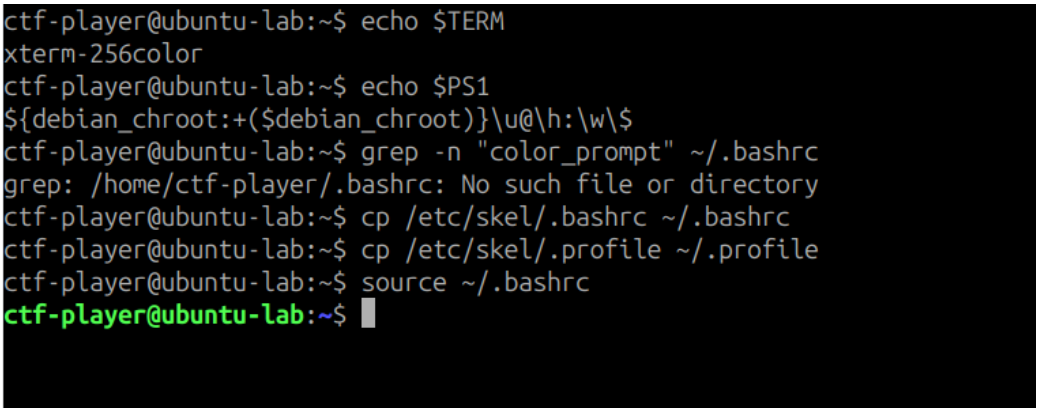


Figure 21: (not sure what this does. I did this so the logged in user shows up in green… and i think i copied the bash config files to ctf-player's home directory.. Double check this)

4. Security measures taken

- Least privilege group assignment.
    - No additional groups were assigned to ctf player user account to maintain least privilege principle.
    - If I were to assign a group like "sudo" then ctf-player will be able to run privileged commands.
- 25 character password
    - Since the Azure VM ip is public and ssh port 22 is also open, everyone on the internet can scan the ip for open ports, therefore if a weak password was used there is a likelihood of bruteforce attack succeeding.
    - To secure against this a 25 character password was used for the ctf-player account which includes numbers and special characters.
    - The possible combinations are 94^25 = 2.3 * 10^49.. therefore it would take 3.8 * 10^29 years even at 1 trillion guesses per second. Meaning bruteforce succeeding is NOT likely anymore.

5. Adding instructions to landing shell session

In when you ssh into a linux machine, the terminal shows a message. This message is stored in bash scripts inside this directory. /etc/update-motd.d/

```
azureuser@ubuntu-lab:~$ cd ..
azureuser@ubuntu-lab:/home$ ls
azureuser  ctf-player
azureuser@ubuntu-lab:/home$ ls /etc/update-motd.d/01-ctf
ls: cannot access '/etc/update-motd.d/01-ctf': No such file or directory
azureuser@ubuntu-lab:/home$ ls /etc
ModemManager                    cron.hourly           group           ldap            mtab                 polkit-1          sensors.d          tmpfiles.d
PackageKit                      cron.monthly          group-          legal           multipath            pollinate         sensors3.conf      ubuntu-advantage
X11                             cron.weekly           grub.d          libaudit.conf   multipath.conf       ppp               services           ucf.conf
adduser.conf                    cron.yearly           gshadow         libblockdev     nanorc               profile           sgml               udev
alternatives                    crontab               gshadow-        libibverbs.d    needrestart          profile.d         shadow             udisks2
apparmor                        cryptsetup-initramfs  gss             libnl-3         netconfig            protocols         shadow-            ufw
apparmor.d                      crypttab              hdparm.conf     locale.alias    netplan              python3           shells             update-manager
apport                          dbus-1                host.conf       locale.conf     network              python3.12        skel               update-motd.d
apt                             debconf.conf          hostname        locale.gen      networkd-dispatcher  rc0.d             sos                update-notifier
bash.bashrc                     debian_version        hosts           localtime       networks             rc1.d             ssh                usb_modeswitch.conf
bash_completion                 default               hosts.allow     logcheck        newt                 rc2.d             ssl                usb_modeswitch.d
bash_completion.d               deluser.conf          hosts.deny      login.defs      nftables.conf        rc3.d             subgid             vconsole.conf
bindresvport.blacklist          depmod.d              init            logrotate.conf  nsswitch.conf        rc4.d             subgid-            vim
binfmt.d                        dhcp                  init.d          logrotate.d     nvme                 rc5.d             subuid             vmware-tools
byobu                           dhcpcd.conf           initramfs-tools lsb-release     opt                  rc6.d             subuid-            vtrgb
ca-certificates                 dpkg                  inputrc         lvm             os-release           rcS.d             sudo.conf          waagent.conf
ca-certificates.conf            e2scrub.conf          iproute2        machine-id      overlayroot.conf     request-key.conf  sudo_logsrvd.conf  wgetrc
ca-certificates.conf.dpkg-old   ec2_version           iscsi           magic           overlayroot.local.conf request-key.d   sudoers            xattr.conf
chrony                          environment           issue           magic.mime      pam.conf             resolv.conf       sudoers.d          xdg
cifs-utils                      ethertypes            issue.net       manpath.config  pam.d                rmt               supercat           xml
cloud                           fstab                 kernel          mdadm           passwd               rpc               sysctl.conf        zsh_command_not_found
console-setup                   fuse.conf             keyutils        mime.types      passwd-              rsyslog.conf      sysctl.d
credstore                       fwupd                 landscape       mke2fs.conf     perl                 rsyslog.d         sysstat
credstore.encrypted             gai.conf              ld.so.cache     modprobe.d      pki                  screenrc          systemd
cron.d                          gnutls                ld.so.conf      modules         plymouth             security          terminfo
cron.daily                      groff                 ld.so.conf.d    modules-load.d  pm                   selinux           timezone
azureuser@ubuntu-lab:/home$ cd /etc
azureuser@ubuntu-lab:/etc$ cd update-motd.d/
azureuser@ubuntu-lab:/etc/update-motd.d$ ls
00-header     50-landscape-sysinfo  85-fwupd              91-contract-ua-esm-status  92-unattended-upgrades  97-overlayroot     98-reboot-required
10-help-text  50-motd-news          90-updates-available  91-release-upgrade        95-hwe-eol              98-fsck-at-reboot
azureuser@ubuntu-lab:/etc/update-motd.d$
```

Figure 22: all the script files in update-motd.d directory

```
ravindu@ravindu-81wb:~$ ssh ctf-player@20.205.124.213
ctf-player@20.205.124.213's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1022-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat Sep 12 13:29:21 UTC 2026

  System load:  0.08              Processes:             118
  Usage of /:   6.8% of 28.02GB   Users logged in:       0
  Memory usage: 34%               IPv4 address for eth0: 172.16.0.4
  Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

   https://ubuntu.com/workshop

1 device has a firmware upgrade available.
Run `fwupdmgr get-upgrades` for more information.


Expanded Security Maintenance for Applications is not enabled.

6 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


*** System restart required ***

1 device has a firmware upgrade available.
Run `fwupdmgr get-upgrades` for more information.
```

Figure 23: how the output of all those scripts appear when a user login into the VM through SSH.

These bash scripts run when a user logs into shell session through ssh. They run certain commands and show information everytime user logs into the VM through ssh. If we want to reset the home directory everytime we login then we can add a script to this directory!

To add a custom login screen message we first:

- make a new script
- Then remove execute permissions from current scripts
  Command used: chmod -x /etc/update-motd.d/*
- Move our file in, give it execute permissions
  Command used: chmod +x /etc/update-motd.d/<filename>

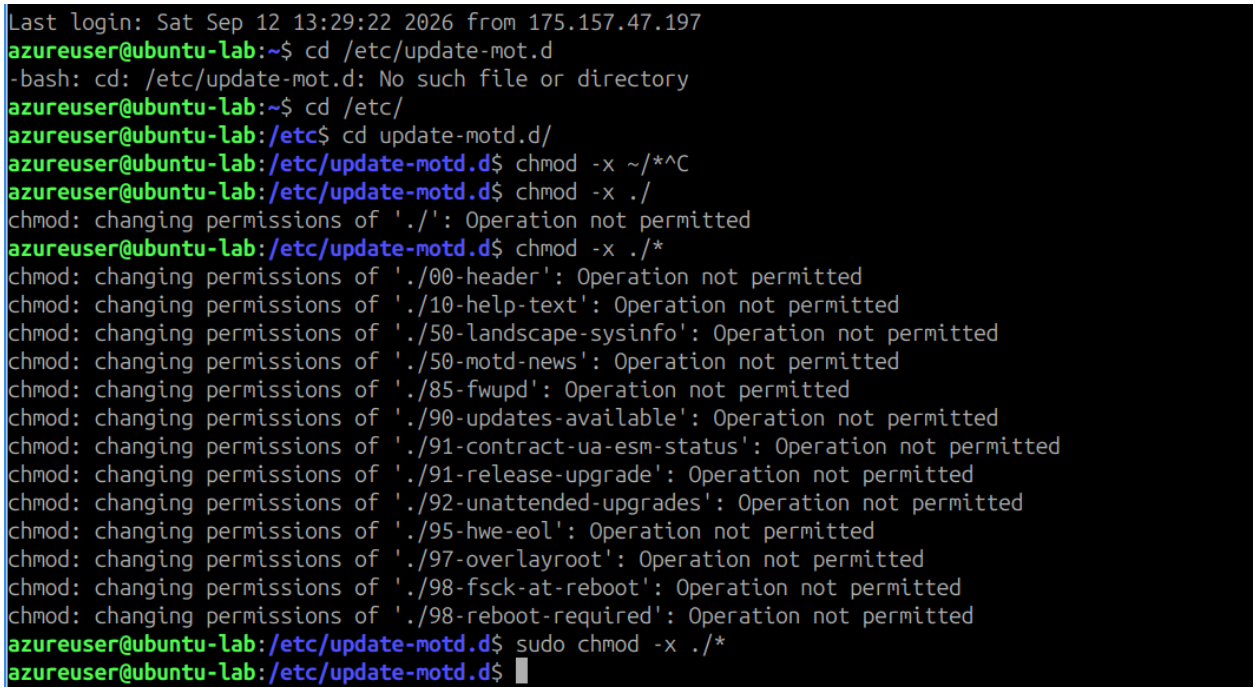


Figure 24: removing permission from all scripts within update-motd.d folder

```
  GNU nano 7.2                                                  /etc/update-motd.d/ctf-welcome
#!/bin/bash
cat <<'EOF'
======================================
 _    _      _ [redacted]
| |  | |    | |
| |  | | ___| | ___ ___  _ __ ___   ___
| |/\| |/ _ \ |/ __/ _ \| '_ ` _ \ / _ \
\  /\  /  __/ | (_| (_) | | | | | |  __/
 \/  \/ \___|_|\___\___/|_| |_| |_|\___|
======================================
EOF
echo
echo "User: $(whoami)"
echo "Current directory: $(pwd)"
echo
echo "The flag you're looking for is in one of"
echo "the files in your home directory!"
echo
echo "Hint: use the 'find' tool to grab the flag..."
echo "Or you can try to read through every file,"
echo "but it won't be an easy task!! ;)"
echo
echo "======================================"
```

Figure 25: contents of the new welcome script

```
chmod: changing permissions of './95-hwe-eol': Operation not permitted
chmod: changing permissions of './97-overlayroot': Operation not permitted
chmod: changing permissions of './98-fsck-at-reboot': Operation not permitted
chmod: changing permissions of './98-reboot-required': Operation not permitted
azureuser@ubuntu-lab:/etc/update-motd.d$ sudo chmod -x ./*
azureuser@ubuntu-lab:/etc/update-motd.d$ nano ctf-welcome
azureuser@ubuntu-lab:/etc/update-motd.d$ sudo nano ctf-welcome
azureuser@ubuntu-lab:/etc/update-motd.d$ sudo nano ctf-welcome
azureuser@ubuntu-lab:/etc/update-motd.d$ sudo nano ctf-welcome
azureuser@ubuntu-lab:/etc/update-motd.d$ chmod +x ctf-welcome
chmod: changing permissions of 'ctf-welcome': Operation not permitted
azureuser@ubuntu-lab:/etc/update-motd.d$ sudo chmod +x ctf-welcome
azureuser@ubuntu-lab:/etc/update-motd.d$
```

Figure 26: adding a new script to update-motd.d folder

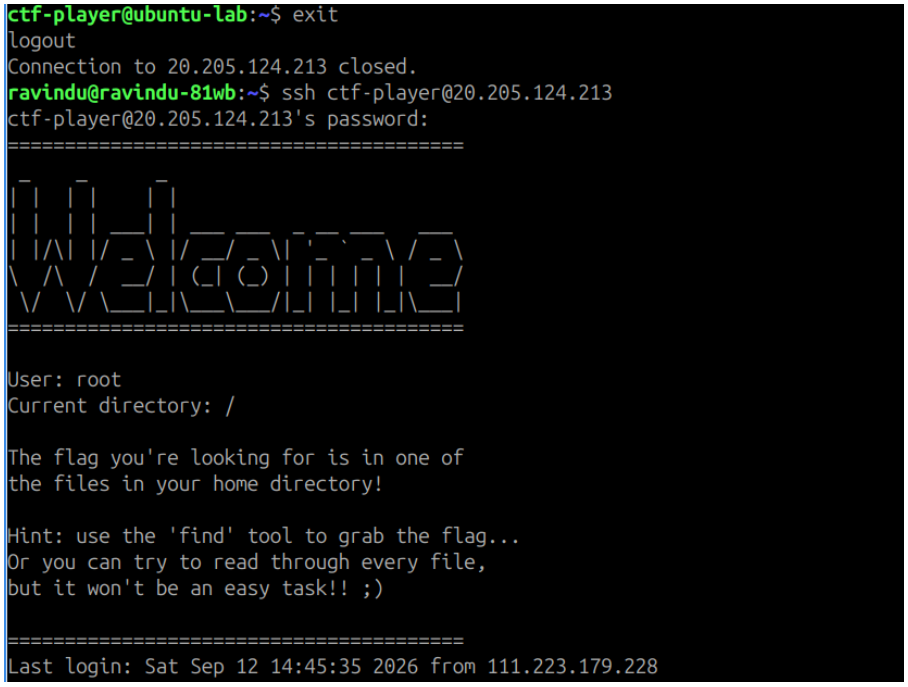


Figure 27: welcome message printed upon ssh login

6. Linux skills learnt

- Difference between ./ and ~/ and / (they represent current directory, home directory of current user and root directory respectively)
- Grep commands were used to find which files were causing the printing of certain outputs.

what grep commands and options (such as -R and -i and -Rni and -nE and -RniE) did we use.

also we used sshd commands with options like -T right what does it do?

what is PAM.d? What is its purpose. is it a niche tool/program. do i need to learn what it is

what does rm -f do. i know rm is remove but why -f

sudo run-parts is a nieche command right, no need to learn what it is?

whats sudo tee? thats new. is it niche and should i ignore it

shell scripting revised... how to echo a line and how to add a variable inside echo line

chmod -x and +x instead of octal numbers

sudo tee /run/motd.dynamic > /dev/null in this line why not 2> /dev/null and instead just >

learnt how to output multi-line echo statements with cat << 'EOF' ---content--- EOF

learnt that echo "| |/\| |/ _ \ |/ __/ _ \| '_ ` _ \ / _ \ " causes problem. in this case better to use cat